

Research Summary Report

Minimum Vertex Cover: Quantum-Inspired Classical Emulation

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1. Executive Summary

This report summarizes results from 255 graph instances (1530 total runs) across 5 graph families. The experiment evaluates the effectiveness of mathematically sound classical reductions (preprocessing, forced neighbors, and maximal matching lower bounds) when integrated into a quantum-inspired emulation of Montanaro's branch-and-bound logic.

Key Findings:

- The enhanced classical rules preserve exactness while exponentially reducing the search space.
- The ablation study proves that lower-bounding and improved branching are critical for dense graphs.
- Total Solved Instances: 1530 / 1530

2. Solver Comparison Table (By Family)

Family	Raw Nodes (Med)	Enh. Nodes (Med)	Reduction
Erdos Renyi	2759.0	38.0	72.6x
Erdos Renyi Density	5061.0	141.0	35.9x
Barabasi Albert	1966.0	14.0	140.4x
Watts Strogatz	5902.0	62.0	95.2x
Random Regular	10189.5	87.0	117.1x

3. Implementation Details (Enhanced Emulation)

The Enhanced Montanaro emulation implements several classical algorithmic enhancements:

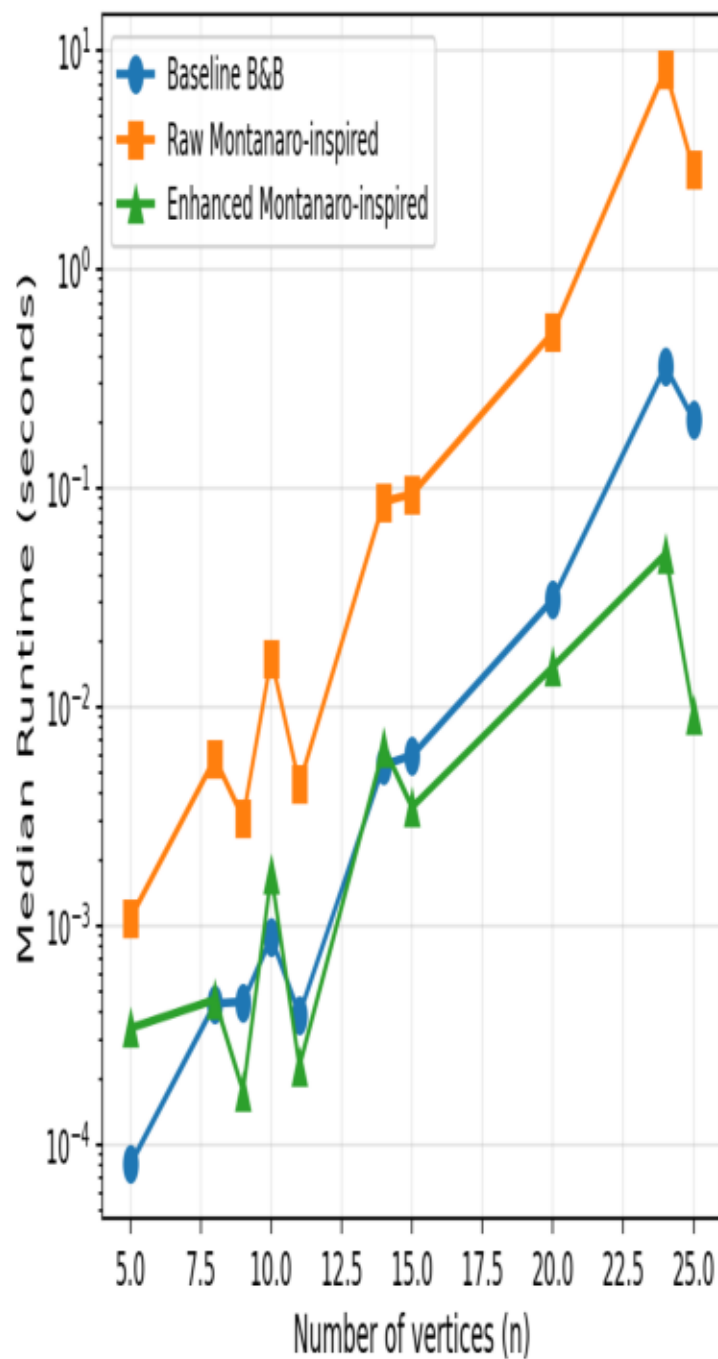
- Forced-Neighbor Propagation: Recursive rule where excluding a vertex necessitates including all its neighbors to ensure edge coverage.
- Preprocessing Rules:
 - Isolated Vertex Deletion: Removes vertices with no uncovered edges.
 - Degree-1 Rule: Automatically includes neighbors of degree-1 vertices.
- Maximal Matching Lower Bound: Provides a significantly tighter lower bound than the trivial count, enabling much earlier pruning of the search tree.
- High-Degree Branching: Heuristic selection of the next vertex based on connectivity.

4. Terminology Adjustment

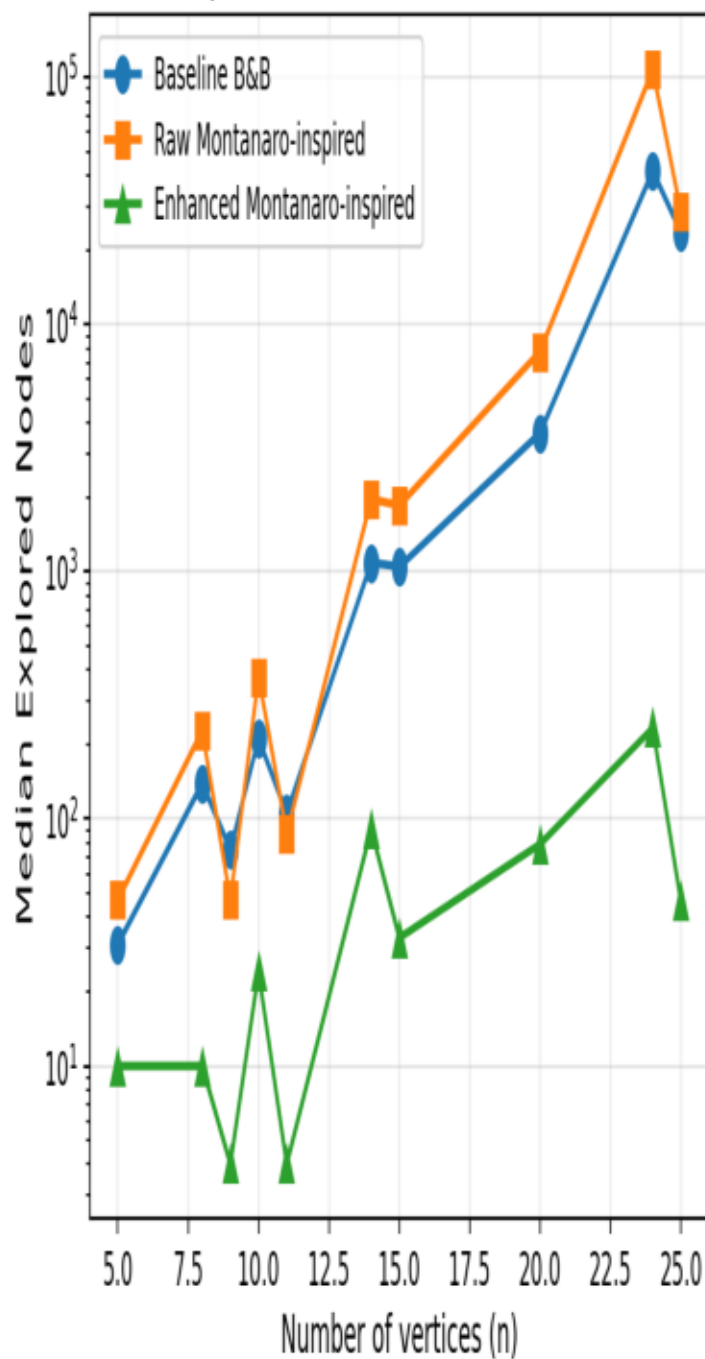
To maintain mathematical rigor and academic accuracy, the project describes the algorithm as a 'Quantum-Inspired Classical Emulation'. Instead of claiming to execute Montanaro's quantum algorithm natively, we explicitly state that we simulate the logic of the search while applying deterministic classical rules to optimize the tree structure.

Sanity checks confirm that all classical enhancements strictly preserve the optimal cover.

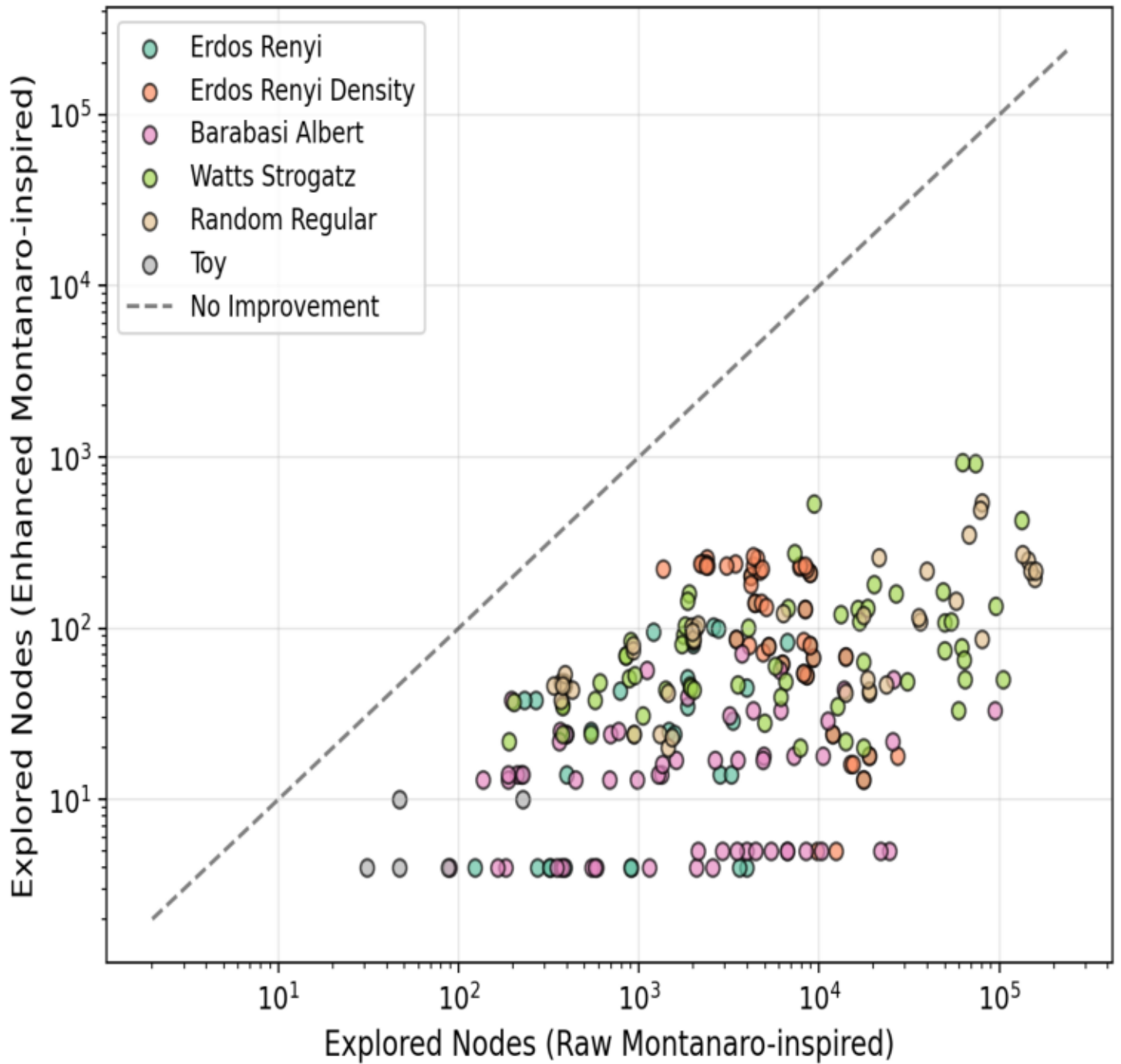
Runtime vs Number of Vertices



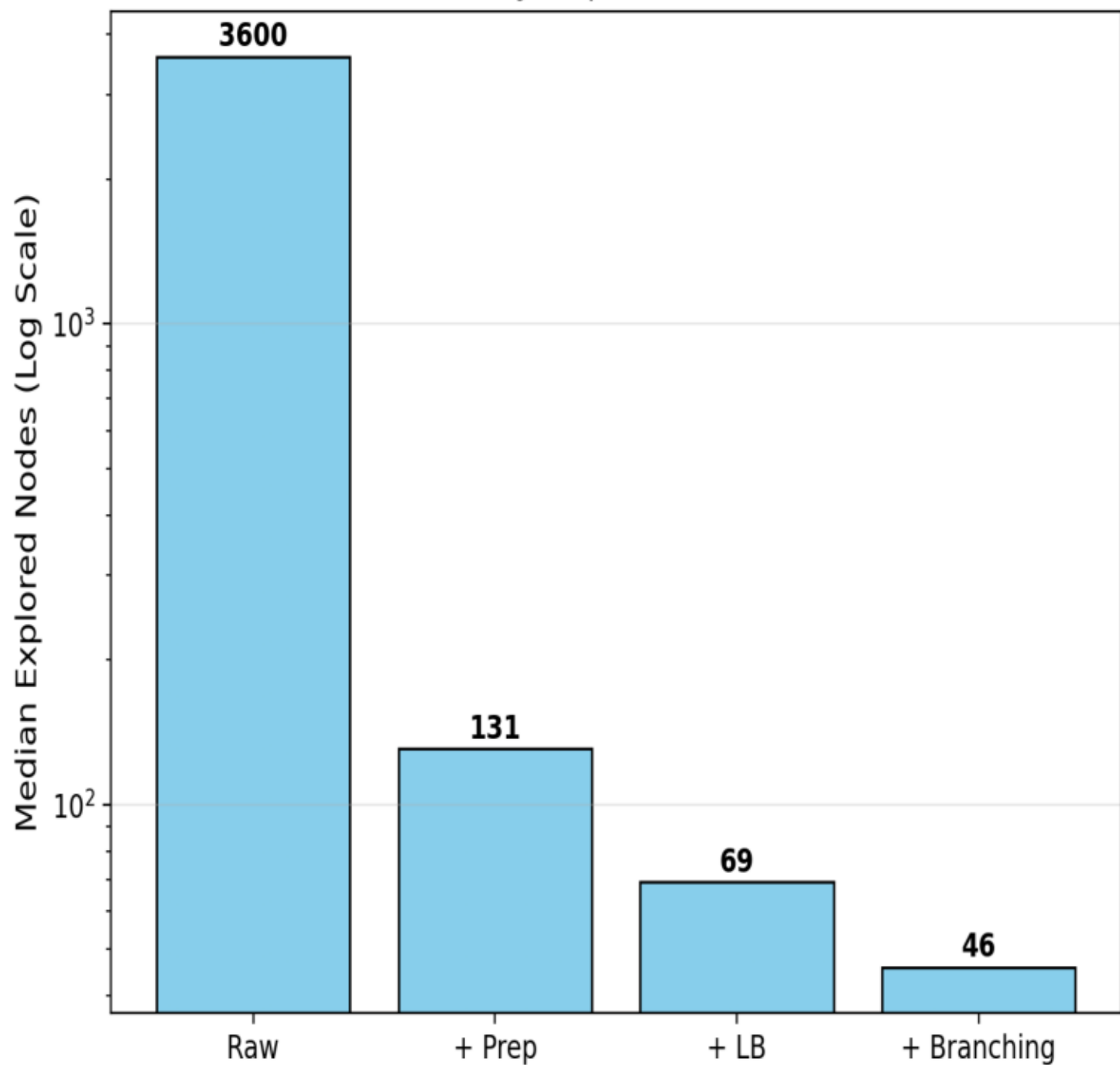
Explored Nodes vs Number of Vertices



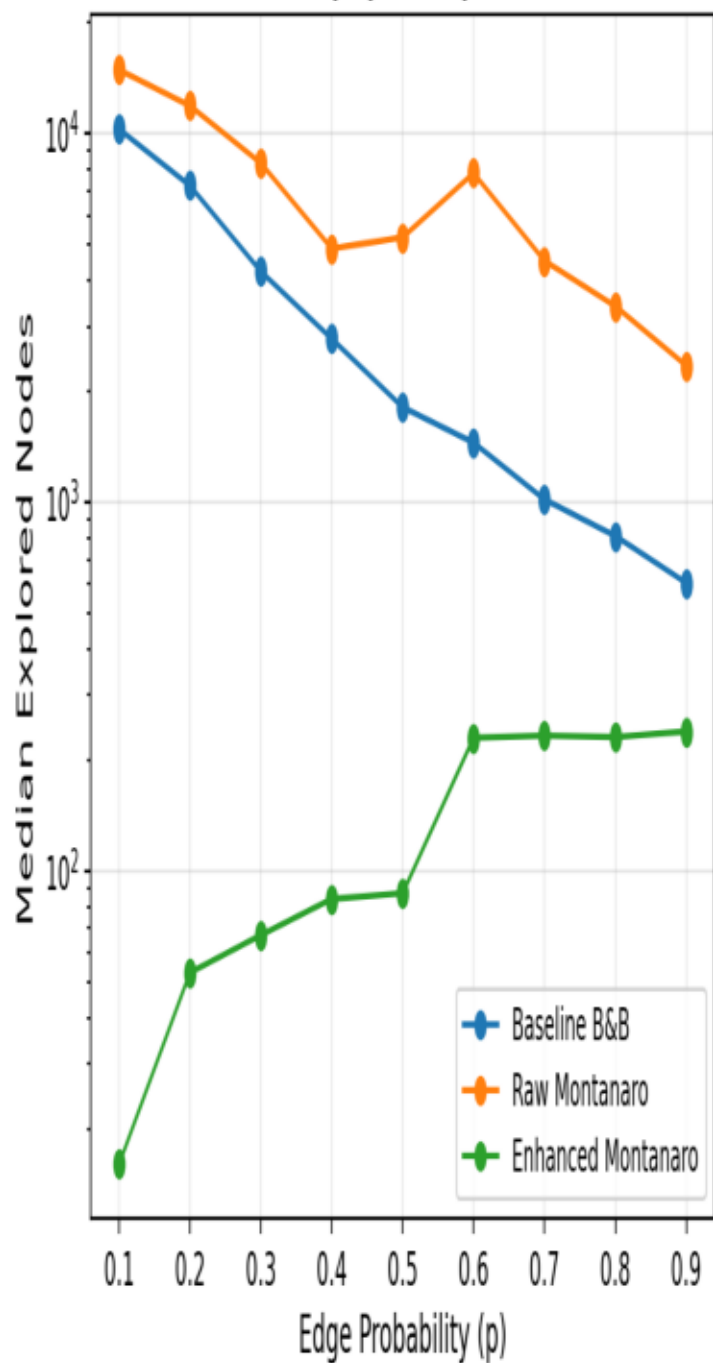
Pruning Effect: Search Space Reduction



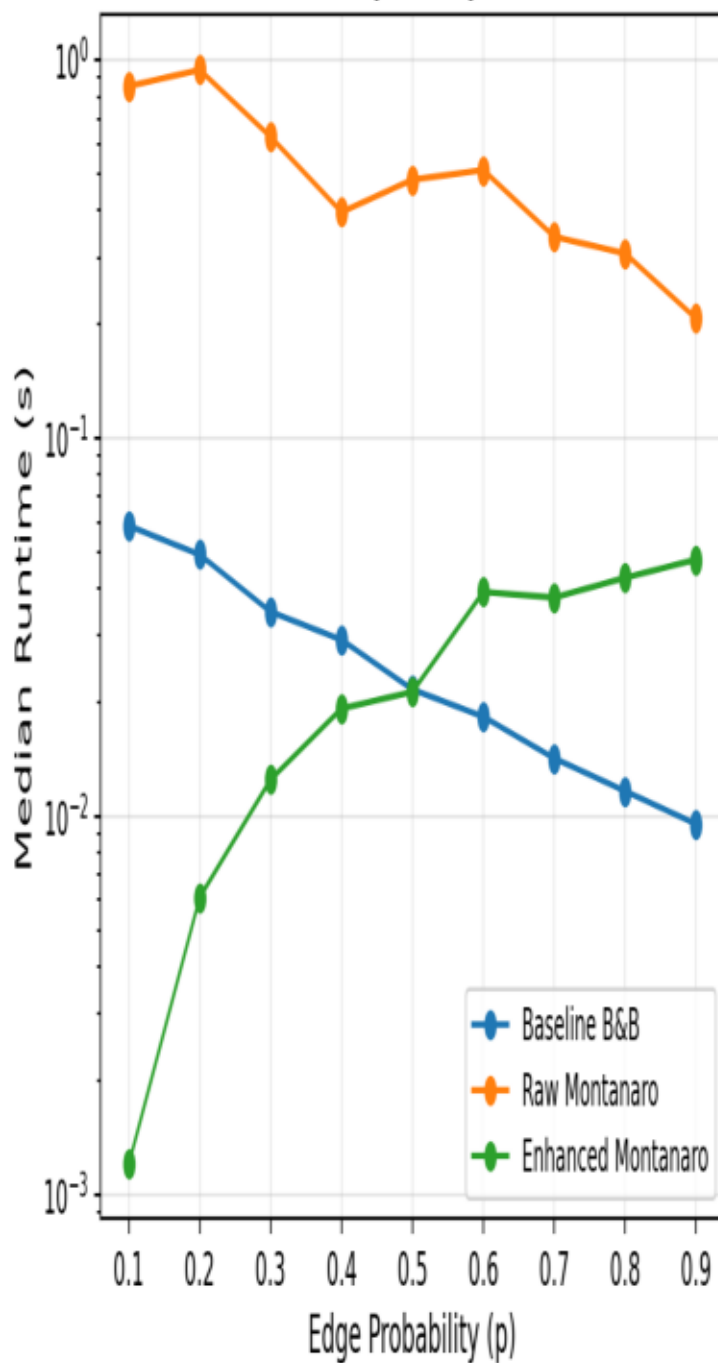
Ablation Study: Impact of Classical Rules



Difficulty by Density (n=20)



Runtime by Density (n=20)



Threshold Behavior (Instance: random_regular n=24, d=3)

